

KRISHNA BALAJI S

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B.Tech in Engineering Physics & M.Tech. in Quantum Science and Technology

Indian Institute of Technology Madras

2021 – 2026

CGPA: 8.68/10

ACHIEVEMENTS

- Secured **All India Rank 833** in JEE Advanced among 150,000 students nationwide.
- Selected as one of 28 Young Research Fellows by IITM among all undergraduates for the 2023-24 cohort.
- Awarded the USEQIP fellowship by IQC, Waterloo for the Summer of 2024.

RESEARCH EXPERIENCE

Master's Thesis: Studying the dynamics of Yb atom in Optical dipole traps

Mar 2025 - Present

Guides: [Dr Athreya Shankar \(IITM\)](#) and [Dr Monika Aidelsburger \(MPQ\)](#)

(Codes not open-source yet, but happy to discuss)

A. Multi-level quantum dynamics in an anharmonic trap

- Developed a Python solver for dissipative quantum dynamics of single-mode multi-level systems by combining grid-based and quantum Monte carlo approaches.
- Studied the effects of state-dependent trapping on atomic qubit and identified experimental regimes where the ^{171}Yb atomic qubit can benefit from imperfect trapping.
- Benchmarked results against the experimental work of [Francesco et al.](#) for ^{171}Yb atom in an optical tweezer, by accurately identifying the non-thermal dynamics of Energy distribution in a trap.
- Currently extending this study to predict particle losses optical lattices during imaging.

B. Multi-mode quantum dynamics using time-dependent variational principle

- Built on the TDVP approach proposed in [Bond et. al](#), [Hackl et. al.](#) to expand our study to three-dimensions.
- Performed quantum-optical calculations to implement a solver that beats QuTip in multi-dimensional laser cooling problems of practical interest.

Master's Internship: Developing a high-flux Yb atomic source

Nov 2025 - Present

Guide: [Dr Monika Aidelsburger \(MPQ\)](#)

- Contributed to design of a vacuum chamber and developed numerical optimization routines for finalizing the ZS, 2DMOT parameters of a high-flux Yb atomic source.
- Involved in everyday lab management including designing setups, ordering AOMs, amplifiers, optics and optomechanics, testing and characterizing current coils, temperature sensors, and permanent magnets.

XRF Spectral Analysis of Lunar Surface ([Github](#))

Oct 2024 - Dec 2024

Indian Space Research Organization

- Analyzed over 5 years of X-Ray Fluorescence data from instruments aboard India's Chandrayaan-2 Lunar Orbiter to generate abundance maps of elements like Al, Mg (1-4.2 keV) on the Lunar surface.
- Modelled the inverse problem by incorporating Matrix effects and Rayleigh scattering using pyXSPEC library.
- Developed a faster x2abundance implementation using dynamic programming and tensorized XRF integrals.

Site-Selective Gating - Rydberg Atom Quantum Simulator Lab ([Key results](#))

May 2024 - Aug 2024

Guide: [Dr Alexandre Cooper-Roy](#), Institute for Quantum Computing, University of Waterloo

- Devised the experimental architecture for a 2D Tweezer array inspired from [Levine Thesis](#) and [Bluvstein et. al.](#)
- Implemented and characterized an Optical Phase Locked Loop (OPLL) between two lasers at qubit frequency.
- Generated a 1D tweezer array by optimizing AOD's diffraction efficiency and aligned it to the objective.

WORK EXPERIENCE

Teaching Assistant

Jan 2025 - May 2025

PH5500 Dynamical Systems

- Teaching assistant for the advanced UG-level Dynamical Systems course under Prof. Arul Lakshminarayan.
- Prepared solutions to problem statements and evaluated examinations across topics spanning stability in Discrete and Continuous time dynamics, Chaos, and Hamiltonian dynamics.
- Contributed to building a commercial UV-Vis-NIR Raman spectrometer system.

ACADEMIC PROJECTS

Review of the Toric code ([Term paper and slides](#))

Jan 2025 - May 2025

PH5842 Advanced Topics in Quantum Computation, IIT Madras

- Prepared in-depth review of the Toric code using the lens of QEC stabilizer formalism inspired by the review from [Dennis, Kitaev, Preskill et. al \(2002\)](#).

Coupling of Neutral atoms via Nanophotonic Cavity ([Term paper and slides](#))

Jan 2025 - May 2025

EE6502 Optical Signal Processing and Quantum Communication, IIT Madras

- Integrated the ideas from [Samputraphoot et. al., PRL \(2020\)](#) and [Bluvstein et. al. Nature \(2024\)](#) to motivate the theoretical description and experimental setup for coupling of two atoms via a nanophotonic cavity.

Protein-folding on a Quantum Computer ([Github](#))

Jan 2025 - May 2025

ID5841 Quantum Computing Lab, IIT Madras

- Reviewed the history of lattice-based models of the protein-folding problem.
- Demonstrated the potential of physically-informed lattice models using IBM's Qiskit quantum compiler.

RELEVANT COURSEWORK

- **Physics:** Atomic Physics, Advanced QM, Quantum Optics, Solid State Physics/Devices
- **Quantum Information:** Quantum Error Correction, Computation, Metrology, Communication
- **Mathematics:** Differential Geometry, Linear Algebra and Convex Optimization, Probability Theory
- **Electrical Engg:** Digital Systems, Analog Electronics
- **Programming/AI:** Deep Neural Networks using Python

TECHNICAL SKILLS

- **Programming Languages:** Python, C++
- **Libraries:** TensorFlow, Pandas, Scikit-Learn, OpenCV, SciPy, PyTorch, Qiskit
- **Frameworks:** Object-Oriented Programming, Functional Programming, Multi-threading, Multiprocessing
- **Software Tools:** COMSOL, MATLAB, Mathematica, CST Microwave Studio, Keysight Genesys, LTSpice

OUTREACH AND VOLUNTEERING

Founder - Quantum Computing Society ([Recordings, slides, other material](#))

Sep 2024 - Present

QCSoc, IIT Madras

- Led efforts to popularize Quantum Computing campus-wide, engaging 700+ students across disciplines.
- Organized sessions on topics ranging from quantum architectures, algorithms, and error correction.
- Presented an intuitive approach to understanding superconducting and neutral-atom architectures using ideas from Magnetic Resonance and high-school-level prerequisites to hundreds of students nationwide.

Mentoring and Outreach

- Delivered a session on Data-Driven Dynamics for nonlinear systems during our institute's Technical fest. ([Github](#))
- Personal mentor to four freshmen, helping them adjust to college life academically and emotionally.

AWARDS AND MISC.

- 3rd place among all Indian institutes in Mimamsa 2023, India's largest science quiz competition.
- Awarded bronze in IEEE's ISIT 2025 Hackathon for a quadratic optimization approach to solve IRS Beamforming.
- National level Chess player with a FIDE rating of 1620.